



Illinois State Water Survey

COOPERATING TECHNICAL PARTNERS (CTP)

PROGRAM MANAGEMENT (PM)

STATEMENT OF WORK (SOW)

PM SOW No. ISWS 22-01 PM SOW

Fiscal Year 2022



FEMA

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Table of Contents

Document Organization and Instructions.....	1
1. Part 1 – Custom Statement of Work Information	2
1.1. Project and Point of Contact Information.....	2
1.2. Tasks and Deliverables to be Completed Under this SOW	3
1.3. Schedule and Performance	20
1.4. Standards	22
1.5. Use of Contractors	23
1.6. Reporting and Performance.....	23
1.7. Privacy and Protection of Personally Identifiable Information	24
2. Part 2 – Available PM Scope Activities	25
2.1. State and Local Business Plans and/or Updates (Required)	26
2.2. Global Program Management.....	26
2.3. Global Outreach for Mapping.....	26
2.4. Training to State, Tribal, Territory, and Local Officials	27
2.5. Mitigation Planning Technical Assistance.....	28
2.6. Staffing.....	28
2.7. Technical Pilot Projects.....	28
2.8. Mentoring and Best Practices.....	28
2.9. Minimal Map Printing.....	29
2.10. Coordinated Needs Management Strategy.....	29
2.11. Programmatic QA/QC Plan.....	30
Authorized Representative Signatures	31
List of Tables.....	32
Acronyms	32

Document Organization and Instructions

This template should be used by Cooperating Technical Partners (CTPs) applying for an award to complete program management (PM) activities.

DISCLAIMER (DO NOT DELETE): *In recognition of the MIP Studies Redesign which was deployed in June 2017, FEMA Region 5 has made an effort to update the accompanying R5 MIP Workbook to align with terminology, projects, purchases and tasks in the redesigned MIP. The Region acknowledges that the signed MAS or SOW may or may not contain the required sections, tables, scope or terminology that is fully aligned with the R5 MIP Workbook. Due to these reasons, the R5 MIP Workbook will be treated as the sole source for creating and obligating projects, purchases and tasks in the redesigned MIP, and must be completed by the CTP.*

This document is organized into two parts to simplify and streamline the Statement of Work (SOW) completion process and improve the usability of the SOW for both the Cooperating Technical Partner (CTP) and Federal Emergency Management Agency (FEMA).

Part 1 provides areas for the CTP to detail custom SOW elements (formatted in tables), along with required elements which cannot be modified by individual CTPs:

- Part 1.1 – 1.3. These sections are where the CTP will provide a narrative on the project and identify the scope elements to be completed under this SOW. Opportunities to clarify or modify the standard descriptions and deliverables on fundable activities (as shown in Part 2) are provided. Additionally, the CTP will indicate the schedule for delivery, leverage, budget, performance measures, etc.
- Part 1.4 – Part 1.7. These sections include standard language the CTP is agreeing to, around standards, use of contractors, reporting and performance, and privacy and protection of personally identifiable information (PII). No inputs or edits may be required by the CTP.

Part 2 of this document provides standard language on available fundable scope activities. If the CTP indicates in Part 1 that they are applying for an individual fundable activity, the CTP is agreeing to scope as written in Part 2 under that subsection unless otherwise noted in Part 1. CTPs have the opportunity to customize this language (by note/deliverable additions, subtractions, etc.) after each scope element in the “Custom Scope Elements.” If the text is accepted as-is, there is no need to copy text from Part 2 into Part 1, it is incorporated by reference

1. Part 1 – Custom Statement of Work Information

In accordance with the CTP Partnership Agreement referenced in Table 1 between **Illinois State Water Survey** (herein referred to as “CTP”) and FEMA, Program Management (PM) Statement of Work (SOW) No. **ISWS 22-01** is as follows:

1.1. Project and Point of Contact Information

Instructions: Complete Table 1 below with the basic project information and point of contact (POC) information for both the CTP and FEMA staff.

Table 1. Project and Point of Contact Information

Information Type	Insert Information
CTP Organization Name:	Illinois State Water Survey
CTP Contractor Working on the activities in this SOW: <i>Optional, only if contractors have already been identified; contractor support may be used for all activities except Staffing and Mentoring, which must be completed by the CTP</i>	NA
CTP Partnership Agreement Date:	September 9, 2013
Period of Performance:	09/01/2022 to 09/30/2023
CTP Project Manager:	Glenn Heistand
FEMA Regional Project Officer (PO): <i>When necessary, additional FEMA assistance should be requested through the FEMA Regional Project Officer</i>	John Wethington 5
FEMA Funding to Complete this PM SOW:	\$85,000

Information Type	Insert Information
CTP Estimated Leverage: <i>Final Leverage dollars or units will be entered as applicable in the Manage Data Development Task Workflow in the Mapping Information Platform (MIP). Leverage data shown here is an estimate of available Leverage at the time the scope is prepared and may be refined throughout the project. See Estimating the Value of Partner Contributions to Flood Mapping Projects “Blue Book” (Blue Book)</i>	NA
Project Team Coordination Activities: <i>Throughout the project, all members of the Project Team will coordinate, as needed, to ensure that activities, products, and deliverables meet FEMA requirements and contain accurate, up-to-date information.</i>	Meetings, teleconferences, and video conferences with FEMA and other Project Team members Monthly Telephone conversations with FEMA and other Project Team members on a scheduled Monthly basis and as needed basis, as required Email as needed As needed

1.2. Tasks and Deliverables to be Completed Under this SOW

CTP must indicate deliverable and leverage details below for activities that are applicable. CTP will also use the R5 MIP Workbook to indicate associated MIP task, grant, cost and schedule details.

1.2.1. NARRATIVE AND AUDIENCE

Instructions: Complete Table 2 below with a high-level narrative of the work to be accomplished under this PM SOW, as well as the intended audience of the project

Table 2. Narrative and Audience

Information Type	Insert Information
SOW Narrative:	The State Business Plan is developed as a joint collaborative document between IDNR-OWR and ISWS, including Global Program Management Activities for Flood Risk Project Mapping Activities. Global Outreach for Mapping tasks is covered in the COMS SOW requested as part of the FFY22 grant. If COMS SOW is not initiated, then Global Outreach for Mapping tasks will need to be added to this PM SOM. Also, if FFY22 COMS SOW is not funded, PM SOW Mentoring and Best Practices task will need to have cost added. Coordinated Needs Management Strategy (CNMS) updates for Illinois are performed by ISWS. Programmatic QA/QC Plans are in place and updated as needed in response to changes in FEMA guidelines and specifications, internal workflow changes, and industry best practices. Project risks identified in the Mapping Activity Statements and CTP Performance Metrics under this grant will be reviewed quarterly and updates provided with the Quarterly Reports submitted through ND Grants. The ISWS Coordinated Hazard and Mapping Program (CHAMP) staff carry out the tasks for FEMA projects. These staff are expected to earn and maintain Certification as a Floodplain Managers (CFM). Maintaining the CFM status of CHAMP staff is part of the management performance and quality assurance protocols for the program. Other specialized certifications to maintain high quality performance standards include Professional Engineer, Geographic Information Systems Professional, and American Institute of Certified Planners. Continuing education credits must be earned annually to maintain certification. Attendance at Illinois Association for Floodplain and Stormwater Management (IAFSM), Illinois Geographic Information Systems Association (ILGISA), Association of State Floodplain Managers (ASFPM), American Planning Association (APA) and other professional conferences, and FEMA training courses, conferences, and webinars provides the opportunity to earn required continuing education credits.
Intended Audience:	State of Illinois, including county and local jurisdictions contained therein., herein referred to as “community(ies)”

1.2.2. PROJECT TASKS AND DELIVERABLES

The following eleven tasks can be accomplished under this PM SOW:

- State and Local Business Plans and/or Updates (**required**)
- Global Program Management Activities (completed under the PM SOW when Recipient is also funded for tasks in the Flood Risk Project Mapping Activity Statement (MAS))
 - Global Outreach for Mapping
 - Training to State, Tribal, Territory, and Local Officials
 - Mitigation Planning Technical Assistance
 - Staffing
 - Technical Pilot Projects
 - Mentoring and Best Practices
 - Minimal Map Printing
- Coordinated Needs Management Strategy (CNMS)
- Programmatic Quality Assurance / Quality Control (QA/QC) Plans

These tasks and their associated deliverables are in listed in the sections below.

Instructions: Please fill out the appropriate sections for the work you plan to complete. Required information includes 1) budget information, 2) identification of deliverables to be provided, and 3) description of the detailed scope elements for each relevant task.

Funding information is required per task. In the leverage table for each task, please indicate the following:

- **A - FEMA Contribution.** This is the funding FEMA is providing to the CTP for the completion of this PM SOW
- **B - Partner Contribution.** Indicate the additional resources required that the CTP will provide to complete the assigned activities for this PM SOW (also known as **Leverage**). Values shall be based on Blue Book values or actual costs where Blue Book values do not exist. The current Blue Book is dated April 2017 and can be downloaded from FEMA's Information Resource Library at https://www.fema.gov/sites/default/files/documents/fema_risk-map_blue-book_2017.pdf.
- **A+B – Total Project Cost** The sum of the above two quantities.

Task 1 - State and Local Business Plans and/or Updates (Required)

This task is **required** as a condition of PM funding – see [Part 2.1](#).

Instructions: Please fill out the required information in Table 3 below.

Table 3. Task 1 – State and Local Business Plans and/or Updates

PM Task	Mark 'X' if task will be done under this SOW	(A) FEMA Contribution	(B) Partner Contribution	(A+B) Total Project Cost			
State and Local Business Plans and/or Updates <i>(required as a condition of PM funding) (see Part 2.1)</i>	☒	\$25,000	\$0	\$25,000			
Deliverable		Mark "X" if deliverable will be done under this task					
Business Plan (required)		☒					
Other: {Insert additional details}		☐					
Custom Scope Elements							
Note, you are agreeing to the scope as written in Part 2.1 unless otherwise modified/noted in this cell. If you accept the text as-is, there is no need to copy the wording here. Only provide additional details here on what you plan to modify. Indicate here if the Community Outreach and Mitigation Strategies (COMS) Engagement Plan and PM Business Plan will be combined as a single deliverable ISWS will collaborate with the Illinois Department of Natural Resources/ Office of Water Resources (IDNR/OWR) a CTP, to prepare a joint business plan for the State of Illinois.							

Task 2 – Global Program Management Activities (to be completed under the PM SOW when Recipient is also funded for tasks in the flood risk project mas)

This task is to be completed under the PM SOW when Recipient is also funded for tasks in the Flood Risk Project MAS – see [Part 2.2](#).

Instructions: Please fill out the required information in Table 4 below.

Table 4. Task 2 – Global Program Management Activities

PM Task	Mark 'X' if task will be done under this SOW	(A) FEMA Contribution	(B) Partner Contribution	(A+B) Total Project Cost			
Global Program Management Activities (to be completed under the PM SOW when Recipient is also funded for tasks in the Flood Risk Project MAS) (see Part 2.2)	☒	\$25,000	\$0	\$25,000			
Deliverable		Mark "X" if deliverable will be done under this task					
Global Program Management Plan		☒					
Other: Updated project risk assessments and CTP Performance Metrics included in quarterly reports.		☒					
Custom Scope Elements							
Note, you are agreeing to the scope as written in Part 2.2 unless otherwise modified/noted in this cell. If you accept the text as-is, there is no need to copy the wording here. ENTER any additional deliverables planned to be developed above. And add relevant specifics that will be completed in this activity in this section. Global Program Management Plan will be integrated in the Business Plan							

Task 3 – Global Outreach for Mapping

Task 3 can be selected under this task if there is no COMS SOW. If a COMS SOW is also completed, the Outreach Plan is required under the Strategic Planning for Community Engagement Task and should not be part of this PM SOW.

Instructions: Please fill out the required information in Table 5 below.

Table 5. Task 3 – Global Outreach for Mapping

PM Task	Mark 'X' if task will be done under this SOW	(A) FEMA Contribution	(B) Partner Contribution	(A+B) Total Project Cost
Global Outreach for Mapping (see Part 2.3)	☐	\$0	\$0	\$0
Deliverable			Mark "X" if deliverable will be done under this task	
Outreach Plan			☐	
Report detailing outreach and coordination activities, including backup or supplemental information used in writing the report			☐	
Business Plan update describing (in detail) the outreach activities			☐	
Updates to CTP's website			☐	
Other: COMS SOW is being requested as part of the FFY22 grant. If COMS SOW is not initiated, then Global Outreach for Mapping tasks will need to be added to this PM SOW.			☒	

PM Task	Mark 'X' if task will be done under this SOW	(A) FEMA Contribution	(B) Partner Contribution	(A+B) Total Project Cost
Custom Scope Elements				
Note, you are agreeing to the scope as written in Part 2.3 unless otherwise modified/noted in this cell. If you accept the text as-is, there is no need to copy the wording here. Only provide additional details here on what you plan to modify. Task 3 – Global Outreach for Mapping will be covered under Communication and Outreach to Communities task in COMS SOW. The ISWS has a history of providing technical data and support to communities, conducting applied research, publishing, and presenting results. In general, ISWS staff take advantage of opportunities to engage the public and provide technical data and support to reduce losses due to flooding. Specific activities envisioned are as follows: The Illinois Association for Floodplain and Stormwater Management (IAFSM) conducts workshops and conferences as well as providing a bi-monthly newsletter. Community officials and engineers who represent communities look to IAFSM for leadership in floodplain management and flood risk mitigation. ISWS staff support IAFSM serving as officers and committee chairs. ISWS staff contribute articles to the newsletter and present at workshops and conferences providing training, education, and updates to communities and engineers on FEMA standards, products, and best practices. ISWS has a booth at the annual Illinois conference to demonstrate FEMA products developed in Illinois. ISWS staff in this leadership role are recognized as experts in flood hazard identification and risk management, developing and maintaining relationships with community leaders and professionals involved with floodplain and stormwater management. ISWS staff participate in the Rapid Assistance Flood Team (RAFT) deployment supported by IAFSM assisting local floodplain managers with post disaster damage assessments. ISWS staff are active in the Illinois Geographic Information Systems Association (ILGISA). ISWS staff serve as officers and assist with annual conferences. ISWS has a booth at conferences to demonstrate FEMA products as well as presenting on FEMA projects. Communities send GIS staff to this conference, and this provides an opportunity to educate them about FEMA GIS products, in particular the DFIRM geodatabase and Flood Risk Database. The Association of State Floodplain Managers (ASFPM) is a national organization of professionals interested in floods, floodplain management and related topics. IAFSM is a state Chapter of ASFPM. ISWS staff support ASFPM activities in a variety of ways including serving as officers, on committees, presenting through hosted webinars and at the national conference. ISWS has a booth at the Annual Illinois Emergency Management Agency Training Summit. ISWS maintains a web site hosting Discovery and other outreach products including reports and geodatabases, studies that are in progress, and providing links to relevant FEMA web sites.				

Task 4 – Training to State and Local Officials

Instructions: Please fill out the required information in Table 6 below.

Table 6. Task 4 – Training to State and Local Officials

PM Task	Mark 'X' if task will be done under this SOW	(A) FEMA Contribution	(B) Partner Contribution	(A+B) Total Project Cost		
Training to State and Local Officials (see Part 2.4)	☐					
Deliverable			Mark "X" if deliverable will be done under this task			
Other: {Insert additional details}			☐			
Custom Scope Elements						
Note, you are agreeing to the scope as written in Part 2.4 unless otherwise modified/noted in this cell. If you accept the text as-is, there is no need to copy the wording here. ENTER deliverables planned to be developed above. And add the specifics on the training and/or audiences that will be completed in this activity in this section. {enter custom scope elements}						

Task 5 – Mitigation Planning Technical Assistance

Instructions: Please fill out the required information in Table 7 below.

Table 7. Task 5 – Mitigation Planning Technical Assistance

PM Task	Mark 'X' if task will be done under this SOW	(A) FEMA Contribution	(B) Partner Contribution	(A+B) Total Project Cost		
Mitigation Planning and Technical Assistance (TA) (see Part 2.5)	☐					
Deliverable			Mark "X" if deliverable will be done under this task			
A report detailing the TA provided, including date(s) of TA and type of assistance and state, tribal, or local community stakeholders supported			☐			
Copies of all technical data provided to local, state, and tribal communities			☐			
Other: {Insert additional details}			☐			
Custom Scope Elements						
Note, you are agreeing to the scope as written in Part 2.5 unless otherwise modified/noted in this cell. If you accept the text as-is, there is no need to copy the wording here. ENTER additional deliverables planned to be developed above. And add the specifics on the technical assistance that will be completed in this activity in this section. {enter custom scope elements}						

Task 6 – Directly Funded Staffing

Instructions: Please fill out the required information in Table 8 below.

If Staffing is the only PM Task to be performed under this SOW, please complete Table 14 and Table 15 below.

Table 8. Task 6 – Directly Funded Staffing

PM Task	Mark 'X' if task will be done under this SOW	(A) FEMA Contribution	(B) Partner Contribution	(A+B) Total Project Cost		
Directly Funded Staffing (see Part 2.6)	☐					
Deliverable			Mark "X" if deliverable will be done under this task			
{Insert number of staff} will attend regional mapping meetings hosted by FEMA regions			☐			
Maintain {Insert number of staff}; potentially utilize {Insert number of staff} current personnel to support CTP program activities			☐			
Other: {Insert additional details}			☐			
Custom Scope Elements						
Note, you are agreeing to the scope as written in Part 2.6 unless otherwise modified/noted in this cell. If you accept the text as-is, there is no need to copy the wording here. However, provide additional details here on what activities and tasks planned for staff to be funded. {enter custom scope elements}						

Task 7 – Technical Pilot Projects

Instructions: Please fill out the required information in Table 9 below.

Table 9. Task 7 – Technical Pilot Projects

PM Task	Mark 'X' if task will be done under this SOW	(A) FEMA Contribution	(B) Partner Contribution	(A+B) Total Project Cost		
Technical Pilot Projects (see Part 2.7)	☐					
Deliverable			Mark "X" if deliverable will be done under this task			
Other: {Insert additional details}			☐			
Custom Scope Elements						
Note, given that pilot activities are by definition new, there are not details of the scope written in Part 2.7. ENTER deliverables planned to be developed above. And add the specifics that will be completed in this activity in this section. {enter custom scope elements}						

Task 8 – Mentoring and Best Practices

Instructions: Please fill out the required information in Table 10 below.

Table 10. Task 8 – Mentoring and Best Practices

PM Task	Mark 'X' if task will be done under this SOW	(A) FEMA Contribution	(B) Partner Contribution	(A+B) Total Project Cost			
Mentoring and Best Practices (see Part 2.8)	☐	\$0	\$0	\$0			
Deliverable		Mark "X" if deliverable will be done under this task					
List of existing or potential CTP entities to be mentored		☐					
Schedule and explanation of Mentoring activities		☐					
Report detailing Mentoring activities provided		☐					
Participant Surveys completed by mentored individuals providing feedback on the quality and benefits of the mentor(s) and Mentoring activities		☐					
{Insert #} Best Practice write-ups to be shared nationally on the CTP Collaboration Center, on the CTP Mentoring Network (via Basecamp) and with FEMA Regional PO		☐					
Other: {Insert additional details}		☒					
Custom Scope Elements							
Note, you are agreeing to the scope as written in Part 2.8 unless otherwise modified/noted in this cell. If you accept the text as-is, there is no need to copy the wording here. Only provide additional details here on what you plan to modify. If FFY22 COMS SOW is not funded, this PM SOW Mentoring and Best Practices task will need to have cost added.							

Task 9 – Minimal Map Printing

Instructions: Please fill out the required information in Table 11 below.

Table 11. Task 9 – Minimal Map Printing

PM Task	Mark 'X' if task will be done under this SOW	(A) FEMA Contribution	(B) Partner Contribution	(A+B) Total Project Cost		
Minimal Map Printing (see Part 2.9)	☐					
Deliverable			Mark "X" if deliverable will be done under this task			
Quarterly report detailing the map printing activities, including recipient community/individual, panel number, number of copies, and total associated cost			☐			
Other: {Insert additional details}			☐			
Custom Scope Elements						
Note, you are agreeing to the scope as written in Part 2.9 unless otherwise modified/noted in this cell. If you accept the text as-is, there is no need to copy the wording here. Only provide additional details here on what you plan to modify. {enter custom scope elements}						

Task 10 – Coordinated Needs Management Strategy

Instructions: Please fill out the required information in Table 12 below.

Table 12. Task 10 – Coordinated Needs Management Strategy

PM Task	Mark 'X' if task will be done under this SOW	(A) FEMA Contribution	(B) Partner Contribution	(A+B) Total Project Cost			
Coordinated Needs Management Strategy (CNMS) (see Part 2.10)	☒	\$25,000	\$0	\$25,000			
Deliverable		Mark "X" if deliverable will be done under this task					
Interim deliverables for the mapping project along its life cycle (funded, Preliminary, Letter of Final Determination (LFD))		☒					
Supporting documentation for new determinations		☐					
A self-certification document outlining the funded scope and comments per region discretion		☐					
A self-certified CNMS spatial database using the CNMS QC tool		☐					
Other: QC and Incorporation of Validation Assessments		☒					
Custom Scope Elements							
Note, you are agreeing to the scope as written in Part 2.10 unless otherwise modified/noted in this cell. If you accept the text as-is, there is no need to copy the wording here. Only provide additional details here on what you plan to modify. {enter custom scope elements}							

Task 11 – Programmatic Quality Assurance/Quality Control Plans

Instructions: Please fill out the required information in Table 13 below.

Table 13. Task 11 – Programmatic QA/QC Plans

PM Task	Mark 'X' if task will be done under this SOW	(A) FEMA Contribution	(B) Partner Contribution	(A+B) Total Project Cost			
Programmatic Quality Assurance / Quality Control (QA/QC) Plans (see Part 2.11)	☒	\$10,000	\$0	\$10,000			
Deliverable		Mark "X" if deliverable will be done under this task					
Programmatic QA/QC Plan, including any updates		☒					
Quarterly report detailing the QA/QC activities completed for flood mapping products		☐					
Corrective Action Plan(s), as needed		☐					
Other: {Insert additional details}		☐					
Custom Scope Elements							
Note, you are agreeing to the scope as written in Part 2.11 unless otherwise modified/noted in this cell. If you accept the text as-is, there is no need to copy the wording here. Only provide additional details here on what you plan to modify. ISWS QA/QC Plans and Checklists Professional development activities to support staff Certified Floodplain Manager (CFM) credentials are important to maintain up-to-date knowledge of the FEMA program, guidelines and specifications, and appropriate QA/QC plans and checklists.							

1.2.3. PERCENTAGE OF TIME SPENT ON TASKS OR ACTIVITIES

Instructions: Table 14 and Table 15 are only required if Staffing is the only PM Task to be performed under this SOW (other than the State/Local Business Plan, which is required). If the CTP is not performing Staffing or is performing multiple activities which include Staffing, these tables may not be required. Coordinate with your FEMA POC on any additional applicability.

Note: Assume each staff member works 100% of their day on this project. This represents percentage of total time on the project, not the hours spent.

Table 14. Percentage of Time Spent on Tasks for Funded Staff

Task	Funded Staff Member 1	Funded Staff Member 2	Funded Staff Member 3	Funded Staff Member 4	Brief Work Description Across All Funded Staff
Program Management	%	%	%	%	
Outreach for Mapping	%	%	%	%	
Training to State, Tribal, and Local Officials	%	%	%	%	
Mitigation Planning Technical Assistance	%	%	%	%	
Mentoring and Best Practices	%	%	%	%	
{Insert additional activities}	%	%	%	%	
Total (not to exceed 100 percent per employee)					

Table 15. Percentage of Time Spent on Tasks for Unfunded Staff

Task	Unfunded Staff Member 1	Unfunded Staff Member 2	Unfunded Staff Member 3	Unfunded Staff Member 4	Brief Work Description Across All Unfunded Staff
Technical Engineering and Mapping	%	%	%	%	
Hazard Mitigation	%	%	%	%	
Risk Assessment	%	%	%	%	
Outreach	%	%	%	%	

Task	Unfunded Staff Member 1	Unfunded Staff Member 2	Unfunded Staff Member 3	Unfunded Staff Member 4	Brief Work Description Across All Unfunded Staff
Program Management and Overseeing Contracts	%	%	%	%	
{Insert additional activities}	%	%	%	%	
Total (not to exceed 100 percent per employee)					

1.3. Schedule and Performance

Instructions: Insert deliverables for all activities included in this PM SOW in Table 16 below. Examples are provided in italics. Deliverables can be listed individually or grouped into a single deliverable date. Due dates will be negotiated with the FEMA Regional PO.

Table 16. PM Task Deliverables Schedule

SOW Activities	Deliverable	Deliverable Due Date	Submitted To
Business Plan (required)	Business Plan	<i>3 months from Award date</i>	<i>FEMA Regional PO</i>
Global Program Management	Program Management Plan	4 months from Award date	FEMA Regional Project Officer
CNMS	CNMS spatial database updates	Regularly through grant period	Regional CNMS database
Programmatic QA/QC Plan	Programmatic QA/QC Plan	5 months from Award date	FEMA Regional Project Officer

The activities documented in this SOW shall be completed in accordance with Table 16. PM Task Deliverables Schedule. If changes to this schedule are required, the CTP shall coordinate with the FEMA Regional PO and other necessary Mapping Partners in a timely manner. Deliverables must be

uploaded to the MIP unless otherwise approved by the FEMA Regional PO and it is the CTP's responsibility to make sure that final deliverables are stored to the MIP prior to the end of period of performance.

Table 17. Performance Measures Targets

Outcome ¹	Output Measurement ² (with customized Target)	Recorded Unit/Scale
Evidence of continued maintenance, using non-Federal funds, of the processes/systems to support collection, development, evaluation, dissemination, and communication of flood hazard and risk assessment data and mapping.	[100 %] of non-federally funded identify maintenance activities for CTP agreement. 1. Maintenance of equipment and supplies, such as hardware, software, licenses, certifications, etc. that are necessary to complete, review, monitor and report on the work	Should be expressed as either "Achieved" or "Not Achieved" for completing non-federally funded maintenance activities for the CTP agreement.
Advancement of program metrics and/or accomplishment of project performance measures, captured as Earned Value	Report on Schedule Performance Index (SPI) and Cost Performance Index (CPI). Must be between 0.92 and 1.08 . If it is not, provide information on what is being done to correct the problem.	SPI Ratio (Budgeted Cost of Work Performed/Budgeted Cost of Work Scheduled [planned]) CPI ratio (Budgeted Cost Work Performed/ Actual Cost Work Performed)
Continued maintenance (using non-federal funds)	0.5 FTE non-federal support. See Business Plan for activities.	Should be expressed as either "Achieved" or "Not Achieved".

¹ An outcome is an observable and measurable change of knowledge, behavior, skills, and/or efficiency due to CTP project.

² An output is a direct, specific, & quantifiable product of CTP activities that lead to /indicate success of the intended outcome, expressed in units of measure that enable quantifiable recording of performance

1.4. Standards

The standards relevant to this SOW are presented in FEMA Policy 204-078-1 Standards for Flood Risk Analysis and Mapping, Revision 12, dated November 2021 located on FEMA's website at <https://www.fema.gov/flood-maps/guidance-reports/guidelines-standards/standards-flood-risk-analysis-and-mapping-public-review>. This Policy supersedes all previous standards included in the *Guidelines and Specifications for Flood Hazard Mapping Partners*, including all related appendices and Procedure Memorandums. Additional information, along with links to guidance documents, technical references, templates, and other resources that support these standards, may be found on the FEMA Guidelines and Standards website at <https://www.fema.gov/guidelines-and-standards-flood-risk-analysis-and-mapping>. FEMA reviews standards on an annual basis and the most current version of the policy should be used.

For any Mitigation Planning Technical Assistance activities, coordinate with the FEMA Regional Project Officer to confirm compliance with regional requirements. Additional information is available in FEMA's *Incorporating Mitigation Planning Technical Assistance* guidance document, available on the FEMA Guidelines and Standards website at <https://www.fema.gov/flood-maps/guidance-reports/guidelines-standards/guidance-femas-risk-mapping-assessment-and-planning>.

For CNMS activities, all work shall be performed in accordance with the requirements specified in the most recent *Coordinated Needs Management Strategy (CNMS) Technical Reference* available at <https://www.fema.gov/flood-maps/guidance-reports/guidelines-standards/technical-references-flood-risk-analysis-and-mapping>.

1.5. Use of Contractors

Check applicable statement in Table 18 below.

Table 18. Use of Contractors

Select One	Description of Contractor Options
☐	Contractor support may be used for all activities within this SOW, except staffing and mentoring, which must be completed by the CTP. The CTP will identify the name of the CTP contractor for services used as part of this SOW. The CTP shall ensure that the procurement for all contractors used for this Program Management Activity complies with the requirements of 2 Code of Federal Regulations (CFR) Part 200. Guidance provided in this part includes, but is not limited to, contract administration and record keeping, notification requirements, review procedures, competition, methods of procurement, and cost and pricing analysis. 2 CFR Part 200 may be viewed online at http://www.ecfr.gov/cgi-bin/text-idx?SID=cc011f4fb962e68cb0da4bc91e8fbb43&mc=true&node=pt2.1.200&rgn=div5. Additionally, contractors must not pose a conflict-of-interest issue.
☒	The CTP does not intend to use the services of a contractor for this SOW. No transfer of funds to agencies other than those identified in the approved cooperative agreement application shall be made without prior approval from FEMA. The CTP shall ensure that the procurement for all contractors, if any are used for this SOW, complies with the requirements of 2 CFR Part 200. Guidance provided in this part includes, but is not limited to, contract administration and recordkeeping, notification requirements, review procedures, competition, methods of procurement, and cost and pricing analysis. 2 CFR Part 200 may be viewed online at http://www.ecfr.gov/cgi-bin/text-idx?SID=cc011f4fb962e68cb0da4bc91e8fbb43&mc=true&node=pt2.1.200&rgn=div5.

1.6. Reporting and Performance

Financial Reporting: Because funding has been provided to the CTP by FEMA, financial reporting requirements for the CTP will be in accordance with Cooperative Agreement Funding Opportunity Announcement, Articles of Agreement, or Award Notice for this SOW. The CTP shall also refer to [2 CFR Part 200](#). The CTP shall provide financial reports to the FEMA Regional PO and Assistance Officer in accordance with the terms of the signed Cooperative Agreement for this SOW.

Performance Reporting: Recipients are responsible for providing a signed performance report using the required list of information shown in the NOFO (or and old SF-PPR, if you prefer) on a quarterly basis throughout the period of performance, including partial calendar quarters and periods where no grant award activity occurs. The CTP shall refer to [2 CFR Part 200](#) to obtain minimum requirements for progress reporting. The FEMA Regional PO, as needed, may request additional information on progress.

The CTP will meet with FEMA and/or its contractor(s) as frequently as needed to review the progress of the project in addition to the quarterly financial and status submittals. These meetings may alternate between the FEMA Regional Office, the CTP office, and conference calls as necessary.

The CTP must report performance of the grant in conjunction with the progress reporting. The performance of the CTP is measured by Table 17 Performance Measures Targets. If you are completing a PM project in conjunction with a Flood Risk Project MAS, then you shall use the measures outlined in that MAS based on the 2022 CTP Performance Measures Matrix. Quantitative Targets for performance measures are defined using the 2022 CTP Performance Measures Matrix in conjunction with your FEMA Regional PO and defined in Table 17.

Earned Value Data Entry:

COMS SOW/PM SOW tasks are now tracked in the MIP. Cost and schedule performance measures are defined in this SOW. These measures will be used to monitor partner performance and to determine future funding eligibility. Earned Value data entry involves updating cost, schedule, and performance (physical percent complete) in the MIP by the CTP each month for each assigned task. The CTP will contact the region to obtain additional guidance as needed for updating COMS/PM efforts in the MIP.

1.7. Privacy and Protection of Personally Identifiable Information

Your organizational access to the MIP signifies that you have access to Personally Identifiable Information (PII). As such, please ensure your organization has coordinated with the region so that each user is meeting the requirements with the new Risk Analysis Management Access Request (RAMSAR) process.

Please contact your FEMA Regional PO for more information.

2. Part 2 – Available PM Scope Activities

Note, unless otherwise noted in Tables 3-13 in [Part 1.2.2](#), CTP is required to fulfill all scope required within the tasks described below.

The activities outlined in this SOW will be completed as specified in the Cooperative Agreement Funding Opportunity Announcement, Award Notice, and/or Articles of Agreement. The SOW may be terminated at the option of FEMA or the CTP in accordance with the provisions of the Partnership Agreement. If the SOW is terminated, all products produced to date must be returned and the remaining funding, provided by FEMA for this SOW, from uncompleted activities will be returned to FEMA.

The objective of the PM tasks and activities documented in this SOW is to recognize activities undertaken by CTPs as part of the active process of managing multiple projects. This is broader than individualized project management, which should be covered more directly in any other assigned MASs.

PM tasks and activities cannot result in the production of a Flood Insurance Rate Map (FIRM). Tasks that can be accomplished under this PM SOW are as follows:

- State and Local Business Plans and/or Updates (required)
- Global Program Management Activities (completed under the PM SOW when Recipient is also funded for tasks in the Flood Risk Project MAS)
- Global Outreach for Mapping
- Training to State, Tribal, Territory, and Local Officials
- Mitigation Planning Technical Assistance
- Staffing
- Technical Pilot Projects
- Mentoring and Best Practices
- Minimal Map Printing
- Coordinated Needs Management Strategy
- Programmatic QA/QC Plans

2.1. State and Local Business Plans and/or Updates (Required)

Intent: State and Local Business Plans and/or Updates must be submitted for a partner to receive funding for program management. Plans must document the capabilities and accomplishments of the partner; explain the CTP's vision for implementing or participating in Risk Mapping, Assessment, and Planning (Risk MAP), such as describing how the partner's activities advance the vision, goals, and objectives of Risk MAP (including encouraging communities to take action to mitigate risk); include updates from previous years' activities (if applicable); identify flood hazard mapping needs based on physical, climatological, or engineering methodology changes and document these mapping needs to be included in the CNMS Geographic Information System (GIS) data model; and provide recommendations to FEMA regarding future Risk MAP projects within the state or local jurisdiction.

CTPs who also receive an award for the COMS SOW may combine the Business Plan requirements of the PM and COMS awards into a single deliverable, with approval from the FEMA Regional PO. The combined Business Plan must include all required elements of the PM Business Plan and the COMS Business Plan.

2.2. Global Program Management

(Completed under the PM SOW when Recipient is also funded for tasks in the Flood Risk Project MAS)

Intent: Program management is the active process of managing multiple related projects that need to meet or exceed predefined performance metrics. Specific metrics are defined on a region-by-region basis, and it is recommended to include and/or reference specific relevant metrics as appropriate in this document. Efforts across a program will be aligned and integrated toward the accomplishment of Risk MAP goals.

PM activities will typically occur in the areas of integration, scope, schedule, cost, quality, human resources (staffing, training, resource enablement), communication, risk, and/or procurement. Some efforts may extend beyond the scope of work defined in the SOW or specific project MAS.

The CTP will work with the FEMA Regional PO during the initiation of this activity to determine a PM Plan for implementation.

2.3. Global Outreach for Mapping

If doing a COMS SOW as well, this activity should be placed in COMS and removed/ referenced here.

Intent: The Outreach project or activities for a PM SOW can best be understood as a process that enhances the understanding of the overall National Flood Insurance Program (NFIP) flood mapping program, including flood risks and hazard identification. This task does not include the Outreach activities for a specific mapping project that begins during the project Discovery phase and continues through the map production and post- preliminary phases.

Note: Communication and Outreach activities described in this task are meant to be supplemental or complementary efforts to those identified in the Community Engagement tasks in the Flood Risk Project MAS. CTPs and the FEMA region are responsible for confirming no duplication of effort in other awards (grants, cooperative agreements, interagency agreements and contracts).

The overarching goal for outreach is to create a climate of understanding and ownership of the mapping process at the state, tribal, and local levels. Well-planned outreach activities can ensure a more positive interaction throughout the Risk MAP process. These outreach activities can also assist FEMA and other members of the Project Management Team (PMT) in responding to Congressional inquiries. The CTP plans to continue Outreach activities to fulfill its strategy to educate communities on the necessary standards and benefits of developing better flood risk information.

The CTP will work with the FEMA Regional Office during the initiation of this activity to determine or understand the Outreach Plan. The FEMA Regional Office will have access to many outreach tools that have been developed for this process that can be utilized or customized. All communication with local governments will be done in accordance with 44 CFR Part 66.

The Mapping Partner shall notify FEMA and all applicable parties of all meetings with community officials at least two weeks prior to the meeting (with as much notice as possible). FEMA and/or its contractor may or may not attend the community meetings.

2.4. Training to State, Tribal, Territory, and Local Officials

Intent: Develop and provide technical training to state, tribal, territory, and local officials throughout the course of a flood risk project. Training can be provided at any time during the flood risk project, and it will include a series of training activities over the course of a flood risk project.

If the CTP elects to perform Training to State, Tribal, and Local Officials, the CTP will coordinate and/or administer training for a defined Community and/or Individual on specific topics. The CTP will:

- Determine target audience;
- Advertise to and confirm training participants;
- Determine training facility;
- Provide training materials;
- Provide training instructors;
- Provide list of participants and evaluations to FEMA;
- Follow up with participants on unresolved issues.

2.5. Mitigation Planning Technical Assistance

Intent: Develop and disseminate products and materials to support states, tribes, and local jurisdictions to develop, evaluate, update, and implement their mitigation plans and strategies. Technical Assistance for Mitigation Planning provided through Risk MAP will focus on building a community's capability to plan for and reduce risk. Technical Assistance will encourage hazard mitigation plan implementation and advance community hazard mitigation actions through the Mitigation Planning Process and Risk MAP projects. The following steps are emphasized:

- Incorporating new flood hazard and risk information;
- Updating and refining mitigation strategies, especially as related to new flood hazard/risk information;
- Training mitigation planning teams;
- Incorporating mitigation into existing community plans, programs, and policies.

This task will be used to provide state and local officials with technical assistance for achieving mitigation actions. This task cannot be used to fund the creation or update of a Hazard Mitigation Plan. This task cannot fund an activity that is already funded through another federal grant (including the COMS SOW) and will not duplicate assistance available to any community engaged in a Risk MAP project or a Hazard Mitigation Assistance (HMA) planning or project grant.

2.6. Staffing

Intent: Provide staff to support Risk MAP Program Management activities. Contractor support may not be used for staffing and mentoring for this SOW. Staffing and mentoring must be completed by the CTP.

2.7. Technical Pilot Projects

Intent: *(As defined by and negotiated with the FEMA Regional Office and approved in coordination with FEMA Headquarters (HQ). Please note that the Letter of Map Revision (LOMR) Review Partner Pilot has its own unique MAS. This PM SOW must not be used for the LOMR Review Partner Pilot Activity.)*

2.8. Mentoring and Best Practices

Intent: Share CTP program experience and related information with peer participants regarding best practices and process improvements.

For this agreement, the CTP shall submit at least one written Best Practice to its FEMA Regional PO detailing something that was innovative, cost saving, enhanced community engagement, etc. to be loaded to the CTP Collaboration Center and CTP Mentoring Network (via Basecamp). If this CTP has a

COMS SOW as well, this Best Practice can be done in collaboration with the requirement in that agreement.

2.9. Minimal Map Printing

Intent: Print copies of maps for individuals within the CTP's jurisdiction. Funding for this activity must not be covered under another FEMA grant program and will not exceed \$5,000.

2.10. Coordinated Needs Management Strategy

Intent: CNMS is a spatial database that each FEMA Region Service Center (RSC) maintains as representation of FEMA's mapped floodplain inventory and a reporting system for New Valid Updated Engineering (NVUE). CNMS is maintained and updated for LOMRs, Discovery, Digital Flood Insurance Rate Map (DFIRM) mapping projects, and for expiring CNMS validation miles. The mapping partner will need to work and coordinate with their respective FEMA RSC to make sure the database is updated in accordance with FEMA specifications.

(The following two optional paragraphs should be identified in the Flood Risk Project MAS. If for some reason a CTP is only doing the CNMS and not the actual discovery or mapping project, this must be coordinated in advance with the FEMA region and explained in this section.)

For this SOW, the CTP will evaluate and update the CNMS database for request area(s) within the **Discovery project** footprint after the meeting. This includes stream reaches identified for study/restudy and any areas with remaining needs and/or requests as appropriate. The mapping partner needs to resolve any discrepancies within CNMS found at Discovery working with the FEMA RSC before the final deliverable.

For this SOW, the CTP will evaluate and update the CNMS database for all studies impacted by the **mapping project** for new or updated studies. This will require updates and deliverables to the FEMA RSC when the project is funded for Discovery, DFIRM production, Preliminary Issuance, and LFD Issuance. These are four vital updates needed in CNMS along a project's life cycle. The mapping partner is to use the *CNMS Technical Reference* sections (S_Studies_Ln Discovery and Scoping Phase Updates, S_Studies_Ln Preliminary Issuance Phase Update, and S_Studies_Ln LFD Issuance Phase Update) when updating CNMS. The *CNMS Technical Reference* can be found at <https://www.fema.gov/flood-maps/guidance-reports/guidelines-standards/technical-references-flood-risk-analysis-and-mapping>. At each of the three update points the CNMS database will be submitted to the FEMA RSC for incorporation into the Regional CNMS database. The FEMA RSC will work with the mapping partner to make sure study information is correct.

For this SOW, the CTP will evaluate and update the database for all current fiscal year expiring miles **and/or unknown miles** as of a specified year(s) for evaluation. Expiring Miles are defined as any miles with a VALIDATION_STATUS of Valid or Unknown and without a STUDY_TYPE of Being-Studied where the STATUS_DATE is 5 years or older within that fiscal year. For example, Flood Creek with a status date of October 10, 2020, will "expire" calendar year October 10, 2025, or FY25Q1.

During CNMS assessments of existing effective Zone A studies, Base Level Engineering (BLE) data will support directly Refined Zone A Engineering study (A5) validation. For additional details about the A5 validation process, refer to “Appendix C: Zone A Study Validation Assessment” in the *CNMS Technical Reference*. BLE data prepared in accordance with the BLE Analyses and Mapping Guidance (February 2018) will meet the standards for comparison against effective floodplains as described in the CNMS technical reference.

All identified detailed (AE, AO, AH) reaches will be evaluated using the checks identified in Appendix B: Detailed Study Validation Assessment of the CNMS Technical Reference, and have appropriate documentation provided. The mapping partner can document the validation findings in check-sheets or within a CNMS database by adding additional fields.

As with detailed validation, all reaches must have appropriate documentation provided (see “Appendix B: Detailed Study Validation Assessment” in the *CNMS Technical Reference*). The mapping partner can document the validation findings in check-sheets or within a CNMS database by adding additional fields.

In addition to incorporating new and/or updated studies, as well as the validation checks on expiring miles (if applicable), the mapping partner will update line work to FEMA specifications so that it can be incorporated into the Regional CNMS database. This will include but is not limited to:

- Utilization of topology rules to eliminate erroneous overlaps, multipart features, elimination of lines smaller than Cluster Tolerance, self-overlaps, and self-intersections. Where possible, the mapping partner will strive to eliminate inappropriate dangles and pseudos. Stream network will have connectivity where appropriate.
- Maintenance of all attributes as indicated by the CNMS Technical Reference (using the latest available release as a guide).
- Utilization of the most recent CNMS QC tool (ArcGIS Add-in) with no critical errors found. Any secondary errors will need RSC approval. The ArcGIS add-in tool can be obtained by requesting it from the RSC.

The mapping partner shall maintain an archive of all data submitted for not less than three years past the Grant closing date.

2.11. Programmatic QA/QC Plan

Intent: Develop and implement a programmatic plan for QA and QC of products funded under the Flood Risk Project MAS. Funding for this task does not include performing QA/QC of products, only the development of a plan for QA/QC and the program management aspects associated with implementing and monitoring the effectiveness of the plan. Activities will include the development of Corrective Action Plans and updates to the QA/QC Plan as a result of quality errors identified by the CTP, FEMA, or FEMA Contractors. Funding for this activity must not be covered under another award and will not exceed \$10,000.

Authorized Representative Signatures

Each party has caused this SOW to be executed by its duly authorized representative.



08/26/2022

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Project Manager
Illinois State Water Survey

Date

John Wethington
Regional Project Officer
Federal Emergency Management Agency, Region 5

Date

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Date

List of Tables

Table 1. Project and Point of Contact Information	2
Table 2. Narrative and Audience.....	4
Table 3. Task 1 – State and Local Business Plans and/or Updates.....	6
Table 4. Task 2 – Global Program Management Activities	7
Table 5. Task 3 – Global Outreach for Mapping	8
Table 6. Task 4 – Training to State and Local Officials.....	10
Table 7. Task 5 – Mitigation Planning Technical Assistance	11
Table 8. Task 6 – Directly Funded Staffing.....	12
Table 9. Task 7 – Technical Pilot Projects.....	13
Table 10. Task 8 – Mentoring and Best Practices	14
Table 11. Task 9 – Minimal Map Printing.....	15
Table 12. Task 10 – Coordinated Needs Management Strategy.....	16
Table 13. Task 11 – Programmatic QA/QC Plans	17
Table 14. Percentage of Time Spent on Tasks for Funded Staff.....	19
Table 15. Percentage of Time Spent on Tasks for Unfunded Staff.....	19
Table 16. PM Task Deliverables Schedule.....	20
Table 17. Performance Measures Targets.....	21
Table 18. Use of Contractors.....	23

Acronyms

BLE	Base Level Engineering
CFR	Code of Federal Regulations
CNMS	Coordinated Needs Management Strategy
COMS	Community Outreach and Mitigation Strategies
CPI	Cost Performance Index
CTP	Cooperating Technical Partner

DFIRM	Digital Flood Insurance Rate Map
FEMA	Federal Emergency Management Agency
FIRM	Flood Insurance Rate Map
GIS	Geographic Information System
HQ	Headquarters
HMA	Hazard Mitigation Assistance
LFD	Letter of Final Determination
LOMR	Letter of Map Revision
MAS	Mapping Activity Statement
MIP	Mapping Information Platform
NFIP	National Flood Insurance Program
NOFO	Notice of Funding Opportunity
NVUE	New Valid Updated Engineering
PII	Personally Identifiable Information
PM	Program Management
PMT	Project Management Team
PO	Project Officer
POC	Point of Contact
QA/QC	Quality Assurance/Quality Control
RAMSAR	Risk Analysis Management Access Request
RSC	Region Service Center
Risk MAP	Risk Mapping, Assessment, and Planning
SOW	Statement of Work
SPI	Schedule Performance Index